



LectureLens AI

Product Requirements Document

Real-Time AI Note-Taking for Students with ADHD & Hearing Impairments

1. Executive Summary

LectureLens AI is a real-time lecture companion designed specifically for students with ADHD or hearing impairments. It listens to a lecture as it happens, transcribes the audio into text using Whisper (a state-of-the-art speech-to-text model), and simultaneously uses an AI agent to extract the most important information — Action Items and Key Concepts — displaying them instantly in an AI sidebar. Students no longer have to choose between listening and writing.

Action Item: Any explicit or implicit instruction that requires future execution by the student (assignment, deadline, task, exam preparation instruction).

Key Concept: Any core idea, definition, framework, or principle that helps understand the lecture content but does NOT require action.

Locked product definition: LectureLens is a real-time attention-layer that converts live lecture speech into structured cognitive signals (Key Concepts and Action Items).

Core Problem

Students with ADHD struggle to divide their attention between listening and writing simultaneously — both tasks compete for the same limited cognitive bandwidth. Students with hearing impairments miss words, sentences, or entire explanations when audio is unclear, accented, or fast-paced. Standard note-taking tools (pen & paper, Google Docs, voice memos) require the student to do the hard work themselves — they capture raw audio or typed words, but provide zero intelligence about what actually matters.

The result: students fall behind, miss critical information, feel anxious during lectures, and produce incomplete notes that fail them at exam time. Current 'solutions' like post-lecture recordings or paid human transcriptionists are either too slow (not real-time) or too expensive for most students.

The High-Level Solution

A browser-based app that records live audio, runs Whisper to produce a rolling transcript, and uses a small language model agent to tag, summarize, and surface 'Action Items' and 'Key Concepts' in under 3 seconds — while the lecture is still happening.

🎯 MVP North Star

★ NORTH STAR (LOCKED): $\geq 50\%$ of pilot students report $\geq 50\%$ reduction in manual note-taking effort within 2 weeks of usage. One behavioural truth. Everything else is a supporting diagnostic.

☑ Action Item & Key Concept extraction accuracy $\geq 85\%$ (human-reviewed evaluation set)

☑ DIAGNOSTIC: User trusts sidebar under delay — measured by continued engagement when P95 spikes above 3s (degraded mode sessions). We do not guarantee 3s; we design UX so spikes are tolerable.

☑ DIAGNOSTIC: $\geq 70\%$ return for a second session within Week 2 (early habit signal).

☑ DIAGNOSTIC: WER $\leq 10\%$ on academic English audio (minimum bar for extraction to be meaningful).

2. Users

User Personas

Persona	Who They Are	Core Need
Maya, 20	Undergrad with ADHD, engineering major. Mind wanders in 90-min lectures. Takes messy notes, misses key formulas.	Auto-extracted Key Concepts with timestamps so she can focus on listening.
James, 22	Grad student with moderate hearing loss. Uses hearing aids but misses 15-20% of words in large lecture halls.	Accurate live transcript + highlighted Action Items he would otherwise miss.
Priya, 19	International student, English as 2nd language. Lectures are too fast for her to process and write simultaneously.	Slowed-down comprehension via structured sidebar that organizes what the professor emphasized.
Sam, 21	Non-disabled student in a dense, fast-paced CS lecture. Wants smarter notes without paying for tutoring.	AI-distilled notes so post-lecture review is faster and more effective.

3. Customer & User Needs

Detailed Problem Statement

The dual-task of listening and writing is cognitively expensive for any human. For students with ADHD, the executive function needed to prioritize, filter, and transcribe lecture content while staying focused is severely impaired. For students with hearing loss, the added cognitive load of lip-reading, adjusting hearing aids, or interpreting partial audio makes simultaneous note-taking nearly impossible.

Even when these students use existing tools (recordings, otter.ai, live captions), the output is a raw, unstructured wall of text. There is no layer of intelligence that says: 'This sentence is a Key Concept' or 'The professor just gave you a homework deadline.'

Pain Points — Frequency & Severity

Pain Point	Frequency	Severity
Can't keep up with lecture pace	Every class (daily)	● Critical
Miss action items (deadlines, assignments)	2-3x per week	● Critical
Notes are incomplete or incoherent	Every class	● Critical
Post-lecture recordings take too long to re-listen	Weekly	● High
Can't afford human note-taker or transcription service	Ongoing	● High
Existing tools give raw text, no structure	Every use	● Medium

Constraints

- Accessibility: UI must comply with WCAG 2.1 AA (essential since our users have disabilities).
- Context: Used in lecture halls — must work on laptops with a single microphone, with background noise.
- Environment: Poor WiFi in some classrooms — offline or low-bandwidth fallback needed in MVP+.
- Privacy: Recording lectures may require consent; the app must surface consent reminders clearly.

5. Value Proposition & Messaging

What Problem We Solve — Per Segment

Segment	Value We Deliver
Students with ADHD	Remove the cognitive burden of deciding what to write. The AI watches for you and flags what matters — so you can focus entirely on understanding.
Students with Hearing Loss	Designed to minimize missed deadlines, concepts, and instructions caused by audio dropout or divided attention. The AI sidebar surfaces what matters in real time — without replacing the student's own judgment.
International / ESL Students	Convert the fire hose of fast English speech into organized, digestible concepts — reducing the language processing burden in real time.
General Students	Turn every lecture into a structured study guide automatically, cutting post-lecture review time in half.

Key Capabilities (High-Level)

- Live Audio Transcription — Whisper-powered, streaming, under 2-second lag.
- AI Sidebar — Autonomous agent that tags Action Items and Key Concepts as they are spoken in real time.
- Session Review Mode — Post-lecture view with full transcript, timeline, and extracted highlights. [Note: Live Mode and Review Mode are distinct product modes with different value propositions — see below.]
- Export — Download structured notes as PDF, Markdown, or Notion/Google Docs.

Before vs. After

Before LectureLens	After LectureLens
Frantically typing everything, missing comprehension	Listening fully; sidebar handles capturing — reducing manual note-taking by 40-70% in pilot validation
50 pages of messy raw notes per semester	Clean, structured Key Concepts + Action Items per lecture
Spending 2 hours re-listening to a 60-min recording	10-minute sidebar review immediately after class

Missing assignment deadlines buried in verbal instructions	Action Items flagged in real time with professor's exact words
Expensive human note-takers (\$25-40/hr)	LectureLens at \$9.99/month

Positioning vs. Alternatives

Alternative	Their Gap	Our Edge
Otter.ai / Rev	Raw transcription only — no real-time intelligence layer	AI agent extracts structure while lecture is live
Human note-takers	Expensive, scheduling friction, not scalable	Available instantly, 24/7, at 1/10 the cost
Post-lecture AI summaries (e.g. Notion AI)	Requires waiting until after the lecture — too late	Real-time AI sidebar — decisions happen IN the lecture
Google Live Captions	Caption only, no extraction, no session memory	Full session notes + structured tags + exportable

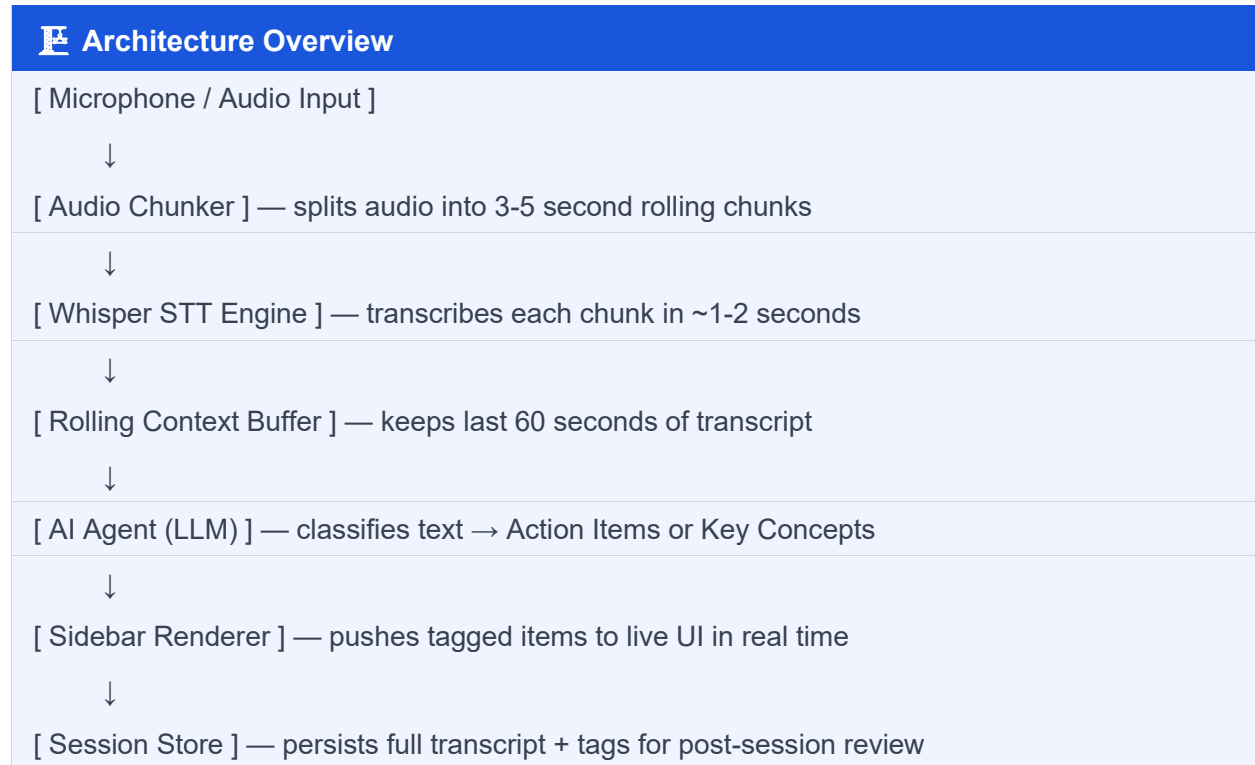
6. Market Opportunity

Metric	Estimate	Source / Notes
Total US college students	~19.5 million	NCES 2024
Students with disabilities (26%)	~5 million	NCES Disability Report
ADHD + hearing impairment subset (~40%)	~2 million primary TAM	CDC + NIDCD data
Willingness to pay \$9.99/mo	~15-20% = 300-400K users	Comparable SaaS conversion rates
Global TAM (higher ed worldwide)	\$2.8B+	EdTech AI tools market 2024
CAGR of AI EdTech market	~18% through 2030	Grand View Research 2024

Why Now

- Before 2023, this product was technically impossible: STT latency was 3–5s per chunk (unusable for real-time), LLM inference took 5–10s for any capable model (GPT-3-class), and small fast LLMs with structured output did not exist. The 2023–2024 window changed three things simultaneously: (1) Whisper Large v3 achieves <10% WER on academic audio with <1.5s latency per 3s chunk. (2) GPT-4o-mini and equivalent models deliver ~0.8s inference with reliable JSON schema adherence. (3) WebSocket + browser MediaRecorder APIs matured enough for stable real-time audio pipelines without native apps. LectureLens is only viable because all three converged in the same 18-month window.
- Small, fast LLMs (GPT-4o-mini) make sub-3-second agent inference economically viable at MVP scale.
- Post-COVID: universities have dramatically expanded disability accommodations budgets and digital tool adoption.
- ADHD diagnosis rates in college-age adults have increased 43% over the past decade (CDC).
- No competitor currently offers a real-time intelligence layer — only raw transcription.

7. System Flow (Architecture)



Step-by-Step Data Flow

- Step 1 — Audio Capture: Browser MediaRecorder API captures mic input continuously.
- Step 2 — Chunking: Audio is sliced into 3-second segments with 0.5s overlap to avoid cutting mid-word.
- Step 3 — Whisper STT: Each chunk is sent to the Whisper API (or local Whisper model). Output: raw transcript text with timestamps.
- Step 4 — Context Buffer: Last 60 seconds of transcript is held in a rolling window to give the LLM context.
- Step 5 — Agent Inference: The LLM agent receives the rolling transcript and a system prompt instructing it to output a JSON payload with {type: 'action_item' | 'key_concept', text: '...', confidence: 0.0-1.0}.
- Step 6 — Sidebar Update: Frontend receives the JSON payload via WebSocket and appends new items to the AI sidebar in real time.
- Step 7 — Session Persistence: After the lecture, the full transcript and all tagged items are saved to the user's session history. Note on real-world non-determinism: the pipeline is asynchronous by design. Whisper API calls use exponential backoff (max 3 retries, 200ms base delay). Audio chunks that fail are queued client-side and replayed on reconnect — preventing data loss without blocking the live stream. LLM agent calls that timeout are dropped (not retried), since stale extraction arriving late is worse than no extraction. The system degrades gracefully: transcript always continues; sidebar may lag or pause.

Latency Budget — Realistic Breakdown (“Real-time” = ≤ 3 s perceived lag, not instantaneous output)

Decision	Insight + Constraint	What We Chose and Why
3-second chunk size (not 1s or 5s)	1s chunks produce too many Whisper partial-sentence errors. 5s chunks feel like a delay, not real-time. Student attention spikes are ~3-4s.	3s with 0.5s overlap. Optimizes STT accuracy AND perceived latency. This is a product decision disguised as an engineering parameter.
Sidebar vs chat interface	Chat requires action (typing). ADHD students cannot context-switch to a chat box mid-lecture without losing the current sentence.	Passive sidebar. Zero interaction required. Information appears without the student doing anything — because attention is the scarce resource.
Streaming vs batch processing	Batch LLM inference on 60-min lectures produces excellent summaries — but 60 minutes after the lecture ends, the student has already taken (bad) notes.	Streaming. The product only has value if it arrives during the lecture — this is the core thesis (articulated in the Key Product Insight section).
Small LLM (GPT-4o-mini) vs large (GPT-4)	GPT-4 inference takes 3-5s. That alone exceeds our entire latency budget. Accuracy gains from larger models don't matter if items arrive after the professor has moved on.	GPT-4o-mini only (MVP). Latency is a hard constraint, not a preference. We accept slightly more brittle classification in exchange for being in-the-moment.

Degraded Mode — UX Expectations

When latency exceeds 3 seconds consistently (e.g., poor campus WiFi), the system enters 'Degraded Mode' automatically. The UX impact must be explicitly defined:

Degraded Trigger	UX Behavior	Student Impact
P95 latency > 3s for 3+ consecutive chunks. P95 latency ≤ 3 seconds means 95% of AI sidebar updates appear	Yellow banner: 'Slow connection — sidebar may lag 5-10s'	Informed, not surprised. Expectations reset.

within 3 seconds from the moment speech is spoken to structured output appearing in the UI. Campus WiFi and Whisper API variability make degraded conditions a common real-world scenario, not an edge case.		
Whisper API timeout (>5s)	Degrade intelligence, not transcription. FIRST: disable AI sidebar entirely and stop LLM inference. Transcript continues via Whisper retry. ONLY if Whisper itself fails (>10s timeout): fallback to browser Web Speech API with clear 'Basic Transcription Mode' label. Rationale: a false Action Item is more harmful than a delayed one. Silence the AI before degrading the transcript.	Lower accuracy transcription — clearly labeled 'Basic Mode'
LLM agent fails (503 / timeout)	Sidebar freezes. Transcript continues normally. Banner: 'AI sidebar paused — full transcript still running'	Student loses AI extraction but keeps raw transcript
Full connectivity loss	Offline banner. Audio buffered locally. Replay attempted on reconnect.	No data lost. Session recoverable.

Sidebar Rendering Rules

RULE 1 — Rate Limiting: Maximum 3 new sidebar items per 30-second window. If more are generated, only the top-K by confidence score are shown.

RULE 2 — Confidence Gate: Items with confidence < 0.65 are held in a hidden 'pending queue' and only promoted if the next chunk confirms them.

RULE 3 — Semantic Deduplication: Before rendering, new items are compared to existing sidebar items. If cosine similarity > 0.85, they are merged (not duplicated).

RULE 4 [V2] — Topic Change Handling: Visual ‘New Topic’ divider on topic shift detection. Requires reliable topic segmentation signal — deferred to V2 to avoid false positives in MVP.

RULE 5 — Action Item Priority: Action Items always jump to the top of the sidebar queue regardless of confidence, since missing a deadline is more costly than clutter.

RULE 6 [V2] — Overflow Handling: Collapsed ‘Show earlier’ toggle when sidebar exceeds 10 items. Deferred to V2; MVP uses simple max cap (Rule 1) instead.

RULE 7 [V2] — Rapid-Fire Pacing: Timed reveal queue (1 per 3s) when >5 items arrive in <10s. Deferred to V2; covered by Rule 1 rate limit in MVP.

8. Data Sources

Data Type	Details
Primary Input	Live microphone audio (WAV / WebM format, 16kHz mono for Whisper optimization)
Supplementary Upload	Optional: student uploads syllabus PDF or lecture slides to give the agent better context
Session Transcript	Full rolling text transcript per session, stored in PostgreSQL with timestamps
Tagged Items	Structured JSON of Action Items + Key Concepts, linked to session ID
User Feedback	Thumbs up/down on each extracted item — used for HITL fine-tuning signal
Model Inputs for RAG	Syllabus + slide content chunked and embedded into a vector store (Pinecone or pgvector)

Storage & Retrieval

- Session transcripts: PostgreSQL (structured, timestamped rows)
- Uploaded documents: S3 / object storage, indexed via embedding pipeline
- Vector store: pgvector (open source, lower cost) for MVP; Pinecone for scale
- User data: Encrypted at rest, no audio stored beyond the active session by default.

9. Model Choice (LLM + Retrieval)

Speech-to-Text: OpenAI Whisper

Attribute	Details
Why Whisper	Best-in-class open-source STT. WER <5% on clean audio, <10% on lecture noise. Free to self-host.
Deployment	Whisper Large v3 via API for MVP (lowest ops burden); self-hosted for cost optimization at scale.
Alternatives Considered	Google Speech-to-Text (higher cost, less open), Azure STT (good but proprietary), AssemblyAI (great but pricier for real-time).
Language Support	99 languages — critical for international student segment.

LLM Agent: GPT-4o-mini (MVP Production) — Claude Haiku (Future Benchmark Only)

Attribute	Details
Why GPT-4o-mini for MVP	Fast (~0.8s per inference), cheap (~\$0.00015/1K tokens), excellent at structured JSON extraction tasks.
Claude Haiku (Future)	Benchmark candidate for future cost-optimization testing at scale only. NOT used in MVP production.
Why NOT GPT-4 or Claude Sonnet	Too slow for real-time use. 3-5 second inference kills the latency requirement.
Prompt Strategy	System prompt defines the extraction schema and classification rules. The prompt instructs the model to output only JSON, with no preamble: {type: "action_item" "key_concept", text: "...", confidence: 0.0-1.0}. Classification logic: Action Items are any obligation with a deadline signal ("due", "submit", "by Friday", "exam"); Key Concepts are named terms, definitions, or ideas the professor emphasizes twice or anchors with "this is important". Few-shot examples (3-5 per category) are included in the system prompt to reduce zero-shot ambiguity. Prompt versioning is tracked; iterations are A/B tested against precision/recall benchmarks on the offline eval set before deployment.

10. Product Scope & Use Cases

Core Features (MVP)

MVP Architecture is progressive, not monolithic. Three explicit tiers: | TRUE MVP (Week 1–2 build): Audio → Whisper → transcript → LLM → label (Action Item or Key Concept) → sidebar. No confidence gating, no deduplication, no rate limiting, no export, no feedback buttons. Show everything. Validate whether the extraction signal has value at all. | MVP+ (Week 3–6, after signal validation): Add confidence gating (0.65 threshold), rate limiting (3 items/30s), semantic deduplication. These exist to prevent cognitive overload — but only after we confirm users trust the raw output. | V1 (post-pilot): Add HITL feedback buttons, export (PDF/Markdown), session review mode. | This layering matters: the PRD describes the full V1 system for completeness, but the pilot validates the True MVP slice only. The MVP validates the core value loop: does real-time structured extraction reduce cognitive load for ADHD and hearing-impaired students?

Feature	Description
Live Transcription	Real-time rolling transcript displayed in main panel. Updates are event-driven: they occur whenever inference completes, typically within a 1–3 second latency window.
AI Sidebar	AI sidebar: live-updating panel showing Action Items (🔴) and Key Concepts (🟡) as they are detected.
Session Recording	Audio captured locally (not uploaded) during session; cleared after session ends.
Post-Session Review	Full transcript + sidebar items in a clean structured layout after lecture ends.
Export	Download session as PDF or Markdown. One-click copy to clipboard.
Feedback Buttons	Thumbs up / thumbs down on each AI sidebar item — feeds HITL improvement loop.
Consent Banner	Clear recording consent reminder shown at session start.

Onboarding & Trust Calibration Strategy

Onboarding is not a tutorial — it is a trust-building system. Without it, ADHD users will either ignore or over-trust the sidebar, both of which are product failure modes.

Session 1 — Calibration Mode: First session runs in “explain mode.” Every sidebar item shows the exact transcript snippet that triggered extraction plus a “why tagged” tooltip (e.g., “Tagged because professor said ‘due Friday’”). Confidence shown visually via color intensity. Student can correct any item in one tap.

Verbatim Anchoring for Action Items: Every Action Item must display the exact transcript phrase that triggered it (e.g., “due Friday” highlighted inline). This is non-negotiable — it is the

primary user-visible hallucination prevention mechanism. Timestamp links allow jump-to-audio for verification in Review Mode.

Sessions 2–4 — Trust Ramp: Explain mode fades as confidence grows. Internal trust score (tracked via sidebar engagement rate, not shown to user) triggers a 1-question check-in if it falls below threshold by Session 3. Primary ICP for onboarding: ADHD students in structured STEM lectures — most predictable lecture format for extraction accuracy, enabling faster trust formation.

User Flow: From Lecture Start to Study Notes

1. Student opens LectureLens in browser → clicks 'Start Session'
2. Optional: uploads syllabus PDF for better context
3. Lecture begins → app records mic audio silently in background
4. Transcript appears in left panel — updates are event-driven as each inference completes (typically 1–3 seconds)
5. AI sidebar populates on the right: Key Concepts appear in blue, Action Items in red
6. Student glances at AI sidebar occasionally — keeps full attention on listening
7. Lecture ends → student clicks 'End Session'
8. Full session review page loads with complete transcript + all extracted items
9. Student downloads PDF notes or exports to Notion / Google Docs
10. Student rates AI sidebar items (thumbs up/down) — 30 seconds of feedback

Input Definition

- Primary Input: Live microphone audio (browser MediaRecorder API)
- Optional Input: PDF (syllabus, slides) uploaded before session starts
- User Input: Thumbs up/down feedback on extracted items post-session

Output Definition

- Live Output: Rolling transcript text (left panel) + AI sidebar items (right panel)
- Post-Session Output: Full transcript page, structured list of Key Concepts and Action Items with timestamps
- Export Output: PDF with session metadata, transcript, and AI sidebar items | Markdown for Notion/Obsidian

MVP vs. Future Scope

Feature	MVP ☒	Future 
Live transcription	☒ Yes	—
AI sidebar — Action Items + Key Concepts	☒ Yes	—
Export PDF/Markdown	☒ Yes	—
HITL feedback buttons	☒ Yes	—
Offline mode	☒ No	V2 — local Whisper + local LLM
Mobile app (iOS/Android)	☒ No	V2
Multi-speaker diarization	☒ No	V3
LMS integration (Canvas, Blackboard)	☒ No	V3
Quiz generation from Key Concepts	☒ No	V3
Shared notes (collaborative)	☒ No	V4

High-Risk Assumptions

- Risk 1: Students will trust AI-extracted notes enough to reduce manual note-taking by 40–70% in MVP validation. Full replacement is a long-term outcome, not an MVP claim. (Validation: pilot study survey at Week 2 and Week 4)
- Risk 2: Whisper performs well enough in noisy lecture halls without dedicated microphone hardware.
- Risk 3: Sub-3-second latency is achievable end-to-end with API-based Whisper + LLM inference.
- Risk 4: Students will actually use the feedback buttons — needed for HITL improvement loop.

11. Risks & Assumptions

This is the #1 existential risk — above latency, above accuracy. If students do not trust the sidebar within their first 2 sessions, they will ignore it permanently. One hallucinated assignment causes immediate churn. ADHD users bifurcate into two failure modes: over-trust (dangerous) or completely ignore (product failure). Trust threshold: $\geq 70\%$ of extracted items must be perceived as correct by the student. Trust is measured by sidebar engagement rate, not just thumbs-up ratio. Mitigation requires an explicit onboarding + trust calibration strategy — see Section on Onboarding.

Risk	Likelihood	Mitigation
Audio quality too poor in large halls	Medium	UI coaching on mic placement; future hardware mic partnership
Latency exceeds 3s in peak usage	Medium	Async processing queue; CDN-edge Whisper deployment
LLM hallucinates Action Items not said	Medium	Confidence threshold filter; HITL feedback flags bad extractions
University bans recording tools	Low-Medium	Consent mode; policy guidance page; opt-in institutional tier. Launch timing risk: students do not adopt new tools mid-semester. MVP pilot must launch at semester start (Week 1–2 only). Trojan horse strategy: position as exam prep / review tool first — lower consent friction — then expand to live lecture use once habit is established.
Students over-rely on AI, disengage from learning	Low	Design nudges to keep transcript visible alongside sidebar
Competitor ships real-time extraction (raw feature parity)	Medium-High	Win via workflow lock-in and classroom habit formation; HITL signals build model lead over time. Feature parity without in-class adoption does not displace us.

12. Evaluation Metrics

Metric	Target & Method
Word Error Rate (WER) [THRESHOLD]	$\leq 10\%$ on academic English. Measured against human-transcribed ground truth corpus.
Precision — Key Concepts [CRITICAL]	$\geq 85\%$: Of all items tagged as Key Concept, 85%+ are confirmed correct by human review.
● Recall — Action Items [CRITICAL — PRIMARY AI METRIC]	$\geq 90\%$ recall on explicit Action Items (those containing explicit deadline signals: "due", "submit", "by Friday", "homework"). Implicit Action Items (e.g., "you'll want to look at this before next week", "this might show up later") are classified separately as [Uncertain — possible action item] with lower confidence weighting. MVP does not target implicit recall; that is V2 scope. Chasing $>90\%$ recall on implicit items at MVP would require contextual reasoning the small LLM tier cannot reliably provide.
F1 Score (combined) [DIAGNOSTIC — secondary]	$\geq 87\%$ across both extraction categories.
Latency (end-to-end) [DIAGNOSTIC — see Section 7 for full budget]	$P50 \leq 1.5s$ $P95 \leq 3s$ $P99 \leq 6s$. Full stage-by-stage breakdown in Section 7.

System Metrics

- Uptime: 99.5% during academic hours (8am–10pm local time)
- P99 API response time: Whisper $< 2s$, LLM agent $< 1.5s$
- Error rate: $< 0.5\%$ of sessions experience a processing failure

Product Metrics

- Week-2 retention $\geq 70\%$ (users return for second session) [DIAGNOSTIC]
- Sessions per user per week ≥ 3 (classroom habit signal) [DIAGNOSTIC]
- Post-session export rate $\geq 40\%$ [DIAGNOSTIC]
- AI sidebar feedback rate $\geq 25\%$ of sessions (HITL signal health) [DIAGNOSTIC]

13. Non-Functional Requirements

Requirement	Specification
Latency	$P50 \leq 1.5s$, $P95 \leq 3s$, $P99 \leq 6s$ (speech $\rightarrow$ AI sidebar). “Real-time” means $\leq 3s$ perceived lag — not instantaneous output. See Section 7 for full per-stage budget.
Scalability	System must handle 10,000 concurrent sessions at launch; auto-scale via cloud functions.
Reliability	99.5% uptime during academic hours. Graceful degradation: transcript continues even if AI sidebar agent fails.
Security	TLS 1.3 for all data in transit. Audio never stored server-side beyond active session. FERPA-compliant data handling.
Privacy	No PII stored beyond email + session metadata. Audio deleted from memory after transcription. GDPR + CCPA compliant.
Accessibility	UI meets WCAG 2.1 AA. High contrast mode. Keyboard-only navigation. Screen reader compatible.

13.2 AI-Specific

Requirement	Specification
Model Performance	Whisper Large v3 for STT. Fast LLM ($\leq 1.5s$ inference) for agent.
Accuracy Thresholds	$WER \leq 10\%$, $Precision \geq 85\%$, $Recall \geq 90\%$ (see Section 12).
Confidence Filtering	Items with confidence < 0.65 are held in a 'pending' state and not shown in sidebar until next chunk confirms.
Monitoring	Real-time latency dashboard. WER tracked weekly via sampled session review. Drift alerts if precision drops $>5\%$ week-over-week.
Model Updates	New Whisper or LLM versions are A/B tested on 10% of traffic before full rollout.

14. Failure Modes

Severity key:  CRITICAL = breaks MVP (hallucinated assignments, missed deadlines).

These cannot ship.  HIGH = degrades experience but recoverable in session.  MEDIUM = tolerable if rare (<5% of sessions). MVP passes QA when zero CRITICAL failures exist in the eval set.

Failure Type	Example	Mitigation
 CRITICAL — Model Failure: Hallucination	Agent tags 'Read pages 40-50' when professor never said it	Confidence threshold + HITL flag + weekly hallucination audits
 CRITICAL — Model Failure: Miss (false negative)	Prof says 'Essay due Friday' — agent doesn't tag it	High recall target ($\geq 90\%$); safety prompts emphasize Action Items
 HIGH — System Failure: Whisper timeout	API call hangs >5 seconds, transcript stops updating	Degrade intelligence first: disable AI sidebar immediately. Retry Whisper (3x, exponential backoff). If Whisper fails entirely: fallback to browser Web Speech API labeled 'Basic Mode' — transcript quality drops but no false Action Items are generated from degraded STT input.
 HIGH — System Failure: WebSocket drop	Live sidebar stops receiving updates mid-lecture	Auto-reconnect with exponential backoff; session recovers on reconnect
 MEDIUM — UX Failure: Sidebar clutter	Too many items tagged → sidebar becomes unreadable	Max 3 new items per 30-second window; deduplication logic
 MEDIUM — UX Failure: Low trust	Student ignores sidebar because early items were wrong	Show confidence indicator; easy correction flow; onboarding calibration session

15. Human-in-the-Loop (HITL)

Feedback Loops

LectureLens improves via a 3-step compounding loop: (1) Student uses real-time AI sidebar during lecture. (2) Student gives thumbs up/down feedback on extracted items. (3) System accumulates labeled (audio → structured output) pairs that improve the prompt and build the future fine-tuning dataset. Human review is used only for weekly sampling — not per-event decisioning.

- **Inline Feedback:** Every AI sidebar item has a 👍 (correct) / 👎 (wrong) button. Feedback is logged with the original transcript context and the extracted item.
- **Weekly Sampling Review:** Daily automated evaluation against a synthetic test set (50 known-label examples with explicit Action Items and Key Concepts). Real-time precision proxy = % of sidebar items downvoted per session (tracked continuously). Weekly: PM reviews 50 random live sessions for qualitative drift. Target iteration cycle: 24–48 hours from signal detection to prompt fix deployment. Weekly PM sampling alone is too slow for an AI product — the synthetic daily eval set is the primary quality gate.
- **Aggregate Signals:** Items consistently rated 👎 are flagged for prompt engineering iteration.

16. Detailed Success Metrics

Metric	Target
Week-4 Retention (WAU/total activated)	$\geq 60\%$
Sessions per active user per week	≥ 3
Post-session export rate	$\geq 40\%$
User satisfaction score (1-5)	≥ 4.2 at 4-week pilot survey
★ NORTH STAR: $\geq 50\%$ reduction in manual note-taking effort within 2 weeks	$\geq 50\%$ report $\geq 50\%$ reduction in note-taking effort (North Star — see Section 1)

Business Success Metrics

Metric	Target
Monthly Recurring Revenue (MRR) at Month 6 [Business model: Freemium for students; premium subscription (\$9.99/month) for unlimited sessions and export; institutional licensing for universities]	\$25,000 (approx. 2,500 paying users)
Customer Acquisition Cost (CAC)	$< \$15$ (campus ambassador + organic)
Lifetime Value (LTV)	$> \$120$ (12-month avg subscription)
LTV:CAC ratio	$> 8:1$
Institutional pilots signed (universities)	3 by end of Month 6

The MVP is declared successful when all of the following are true after a structured 4-week pilot with 20 ADHD STEM students across 2 universities (10 lectures each, 200 total sessions):

Killer Experiment: In a 2-week within-subject trial, students using LectureLens will reduce post-lecture re-listening time by $\geq 50\%$ compared to their baseline (control week without tool).

This is the single most investable signal — if they do not need to re-listen, the in-lecture extraction worked. | North Star Measurement — Behavioral Proxies (not self-report): Primary: post-lecture replay time (logged automatically). Secondary: notes word count vs baseline (keylogging opt-in). Tertiary: Action Item capture rate verified against lecture ground truth. Self-reported “effort reduction” is a supporting signal only — behavioral data does not lie; survey data does.

Structured Validation Experiment: Real-Time vs Post-Lecture

The core product bet — that real-time intervention is better than post-lecture AI summary — must be tested explicitly, not assumed. The pilot should run a within-subject comparison:

Group A (Real-Time): LectureLens running during lecture. AI sidebar active. Student experiences the product as designed.

Group B (Post-Lecture AI): No sidebar during lecture. After class, student receives an equivalently high-quality AI-structured summary (same model, same extraction schema, just delivered 10 minutes after lecture ends) via the review interface.

Group C (Control): No AI assistance. Standard note-taking. Establishes baseline cognitive load and Action Item capture rate.

What each group measures:

- Action Item capture rate: did the student record the correct deadline/task? Verified against ground truth from lecture recording.
- Post-lecture re-listening time: how many minutes did the student spend replaying the lecture? (North Star proxy)
- Perceived cognitive load during lecture: 1-question exit survey — “On a scale of 1-5, how much mental effort did note-taking require during the lecture?”
- Recall test at 48 hours: short quiz on Key Concepts from the lecture. Tests whether in-lecture organization improves retention vs post-lecture summary.

Success Criterion	Target	Why It Matters
★ NORTH STAR: ≥ 50% reduction in manual note-taking effort within 2 weeks	≥ 60%	Core value prop: AI offloads the note-taking task
Students relying primarily on sidebar for post-lecture review	≥ 40%	AI output trusted enough to replace manual notes — key trust signal
Week-2 retention (return for 2nd+ session)	≥ 70%	Early habit formation — if they come back after first session, the product has real pull
Action Item extraction accuracy (human-reviewed)	≥ 85%	Minimum accuracy threshold before product causes harm (missed deadlines)
End-to-end latency (P95)	≤ 3s	Core product promise; above 3s the product feels like a delay not an assistant
Participants who want to continue using after pilot ends	≥ 75%	Strongest signal of product-market fit before open beta launch

18. Competitive Advantage

Why Not Just Use ChatGPT + a Recording App?

The workflow students actually use: record lecture on phone, upload to Whisper/Otter, ask ChatGPT to summarize it, receive a wall of text 90 minutes after class ends. By then, the student has already tried to take notes (badly), missed the assignment deadline mentioned in passing, and spent 30 minutes re-listening.

What LectureLens does differently:

- Latency during cognition: ChatGPT post-hoc processing arrives after the moment of learning has passed. LectureLens surfaces the action item while the professor is still explaining it — when it can actually influence whether the student writes it down or not.
- No workflow embedding: a student using ChatGPT must stop, switch apps, type a prompt, wait, and parse a wall of text. LectureLens requires zero interaction. The AI sidebar is peripheral vision, not a task.
- No attention-aware UI: ChatGPT returns everything. LectureLens returns only what an ADHD student can process in a 3-second glance — max 3 items per 30 seconds, action items always first.
- No real-time prioritization: a recording + summary tells you what was said. LectureLens tells you what to do, during the window when doing it is still possible.

Defensibility Stack

1. LATENCY ENGINEERING — Getting STT + LLM agent under 3 seconds in a real-time loop is a hard engineering problem. It requires specialized audio chunking, context management, and inference optimization that takes months to tune.

2. STRUCTURED EXTRACTION SCHEMA — The exact taxonomy (Action Items vs Key Concepts) and prompt engineering developed through thousands of lectures is a transferable moat.

3. TEMPORAL DATA ADVANTAGE — LectureLens is building the only dataset of real-time cognitive prioritization signals captured during live lectures. Every HITL thumbs-up/down is a labeled data point mapping “what was said in this moment” to “what a student found cognitively significant.” No one else has this. Post-lecture AI tools see the same transcript — but never the real-time signal. This dataset compounds into a proprietary fine-tuning asset that cannot be replicated from public data.

4. DEFAULT LECTURE LAYER — We become embedded inside the student’s semester workflow, not just their note-taking habit. A student who uses LectureLens in Week 1 of a semester has session history, learned patterns, and a calibrated tool by Week 8. Switching means starting over. Workflow presence + institutional procurement = durable moat. HITL signals compound the model advantage, but they are not the moat itself.

5. ACCESSIBILITY-FIRST TRUST — We are built FOR disabled students. Brand trust in this community is earned slowly and lost quickly. Being first means being default.

6. INSTITUTIONAL RELATIONSHIPS — University disability services offices are sticky. Once a school deploys us as an accommodation tool, switching costs are very high.

Why Incumbents Cannot Catch Up

The moat is not just what we are building — it is what incumbents would have to fundamentally re-architect to compete:

Dimension	Incumbents (Otter, Notion AI, Google)	LectureLens
Processing model	Batch-first: process after recording ends	Streaming-first: structured extraction during the lecture
Latency architecture	Not optimized for sub-3s real-time loops	Entire stack optimized around 3s budget (chunking, model tier, RAG gating)
UX design intent	General productivity; not ADHD-specific cognitive load design	Rate limits, confidence gates, dedup, overflow — all designed to minimize sidebar overwhelm
Training feedback loop	None — general models, no domain-specific fine-tuning loop	In-session HITL feedback drives proprietary fine-tuning dataset; compounds weekly

19. Go-To-Market Strategy

Phased Rollout Plan & Revenue Strategy

Phase	Timeline	Focus
MVP / Closed Beta	Month 1-2	20 students with ADHD/hearing impairment at 2 universities. Validate latency, accuracy, UX.
Open Beta	Month 3-4	500 students, self-serve signup. Campus ambassador program. Freemium model (3 sessions free/month).
Paid Launch	Month 5-6	\$9.99/month individual plan. First institutional pilots with disability services offices.
Scale	Month 7-12	Campus partnerships, LMS integrations in roadmap, expand to UK + Canada universities.

Revenue Sequencing: Who Pays First?

Phase 1 (Months 1–4): Free / freemium only. Students have low early willingness-to-pay; forcing payment before trust formation kills adoption. Goal is usage growth and HITL data accumulation, not revenue.

Phase 2 (Months 5–8): Institutional pilots generate first real revenue. Universities pay for accommodations budgets — faster conversion than individual students. These are the actual first-dollar customers.

Phase 3 (Months 9+): B2B becomes the primary revenue engine. Individual subscriptions (\$9.99/month) supplement but do not lead. Long-term moat is institutional contracts + LMS integrations, not consumer SaaS.

Target ICP (Ideal Customer Profile)

- MVP ICP (Primary Beachhead): ADHD students in structured STEM lectures at 4-year US universities. Rationale: STEM lectures follow predictable patterns (definitions, problem sets, deadlines) that maximize extraction accuracy. ADHD is the largest disability segment. This is not the only segment — but it is the only one we optimize for in MVP. All other personas (hearing impairment, ESL, general students) are V1+ expansion targets.
- Institutional: University disability services office with 200+ registered students, existing budget for digital accommodation tools.

Acquisition Strategy

Channel	Tactic
Campus Ambassadors	Partner with 5 students per campus — they demo to classmates, get free subscription + referral commission.
Disability Services Outreach	Direct email to disability directors at top 100 US universities offering free 30-day institutional pilot.

ADHD Community Channels	Reddit (r/ADHD, r/college), Discord servers, CHADD community forums.
Hearing Loss Organizations	Partner with Hearing Loss Association of America (HLAA) — newsletter + webinar.
Content Marketing	YouTube shorts: 'How I take notes with ADHD using AI' — organic search + TikTok.
University Press Releases	Co-author accessibility case studies with partner universities for credibility.

Pilot Strategy

- Pilot 1: 2 universities, 10 students each, 4 weeks. Goal: validate WER, latency, and extraction accuracy in real lecture halls.
- Pilot 2: Partner with a disability services office to offer LectureLens as a formal accommodation option. Measure satisfaction vs. human note-taker.
- Success criteria for pilot: $\geq 75\%$ of pilot students rate the tool 4/5 or higher and want to continue using it.

20. Assumptions & Unknowns

Key System Assumptions

Audio & Environment Assumptions

- Lecture audio is primarily single-speaker (professor dominant)
- Background noise is present but not dominant (> -10 dB SNR typical)
- Students use laptop microphones without external hardware
- Speech is predominantly continuous English academic discourse

Model Behavior Assumptions

- Whisper performance remains stable across lecture domains (STEM, humanities, law)
- LLM maintains consistent structured output under rolling context windows
- Action Item signals are mostly explicit (“homework”, “due”, “assignment”)
- Key Concepts are verbally emphasized by instructors at least once per topic

Product Behavior Assumptions

- Students glance at sidebar intermittently (not continuously)
- Users trust AI suggestions at least 60–70% of the time to reduce cognitive load
- Lecture structure follows topic segmentation (not fully unstructured speech)

Key Unknowns (to validate in pilot)

- How often Action Items are implicit vs explicit in real lectures
- Whether ADHD users tolerate real-time sidebar without cognitive overload
- Real-world variance in Whisper accuracy across noisy lecture halls
- Student trust calibration over time (over-trust vs under-trust risk)
- Whether 3-second latency is “felt real-time” or still distracting

Explicit Non-Assumptions

The system does NOT assume:

- Perfect transcription accuracy
- Stable internet connectivity in classrooms
- Clean topic segmentation in lectures
- Consistent professor speaking pace

21. Model Behavior Risks (LLM + STT Failure Modes)

Whisper (STT) Failure Modes

- Audio clipping → partial sentence loss leading to incorrect extraction context. [Real classroom note: professor movement, overlapping voices, laughter, and domain jargon (math symbols, CS notation) routinely push WER to 15–25% — not edge cases. The 10% WER target assumes typical academic speech. Mitigation: confidence gate suppresses AI extraction during high-noise periods rather than generating low-quality Action Items from corrupted input.]
- Background noise bursts → transient spike in WER
- Accent variability → systematic under-transcription for non-native speakers
- Over-segmentation → splitting sentences incorrectly across chunks

LLM Agent Failure Modes

- Schema overfitting → forces classification even when signal is weak
- Hallucinated structure → generates Action Items not explicitly spoken
- Context drift → misinterprets earlier lecture content in long sessions
- Chunk inconsistency → same concept classified differently across time windows
- Implicit loss → misses references like “this will be on the exam” (expected in MVP; implicit Action Items are classified as [Uncertain] not silently missed)

Cross-System Failure (STT + LLM Interaction)

- Whisper errors cascade into LLM misclassification
- Chunk overlap causes duplicate or conflicting extraction
- Context buffer contamination (old topics influence new ones)

Mitigation Strategy Summary

- Confidence gating (<0.65 suppression)
- Deduplication via semantic similarity threshold (0.85 cosine)
- Rolling window context reset every topic shift
- Post-session reconciliation pass (optional batch refinement)

22. Evaluation Strategy (Offline + Online System)

Offline Evaluation (Pre-Deployment)

Dataset Construction

Sample 100–300 recorded lectures across domains: STEM lectures, humanities lectures, and fast-paced technical talks. Each transcript chunk is annotated for Action Items (binary + timestamp), Key Concepts (multi-label), and implicit vs explicit categorization.

Metrics Computed

- Precision / Recall / F1 per category
- Chunk-level classification accuracy
- Latency distribution (P50 / P95 / P99)
- False positive Action Item rate (critical metric)

Online Evaluation (Production)

Live Metrics Tracking

- User feedback ratio (👍 / 👎 per item)
- Session-level trust score (derived from feedback density)
- Drop-off correlation with sidebar density
- Latency vs engagement correlation

A/B Testing Framework

We continuously test: prompt versions (v1 vs v2 system prompt), chunk sizes (3s vs 5s), confidence thresholds (0.6 vs 0.65 vs 0.7), and sidebar density rules.

Drift Monitoring [V2 Operational Detail]

Alerts: precision drops >5% week-over-week, Action Item recall drops in STEM, or latency P95 exceeds threshold for 3 consecutive days. Full drift system is V2; MVP uses weekly manual sampling review only.

23. Cost / Latency / Accuracy Tradeoff Layer

Core Tradeoff Principle

LectureLens operates under a strict triangle constraint: Low latency ↔ High accuracy ↔ Low cost. Only two can be optimized at once.

Tradeoff Decisions Table

Decision	Choice Made	Why	What We Sacrificed
Chunk size	3 seconds	Optimal latency balance; 5s feels laggy, 1s produces too many partial sentences for Whisper to handle accurately	Slight context loss at sentence boundaries; mitigated by 0.5s overlap
LLM model tier	GPT-4o-mini / Haiku	~0.8s inference fits latency budget; GPT-4 / Sonnet adds 3–5s and exceeds budget	Deeper reasoning; more brittle on edge-case implicit signals
RAG (retrieval)	Optional, latency-gated	Retrieval must complete in <300ms P95 to be used; skipped automatically if exceeded	~5–8% precision gain when course materials are uploaded
Whisper deployment	API for MVP; self-host at scale	API eliminates infra ops burden in MVP phase; self-host needed for unit economics at scale (>50K sessions/day)	Higher per-minute cost in MVP; latency variance when self-hosting later
Sidebar density	Max 3 items / 30s	Cognitive overload is the primary ADHD risk; too many items defeats the product's purpose	Higher miss rate on lower-confidence Key Concepts; they queue for next window

24. What We Are NOT Building in v1

Explicitly Out of Scope

NLP / AI Features Not Included

- Speaker diarization (multi-speaker separation)
- Visual slide understanding (OCR + multimodal parsing)
- Mathematical equation interpretation
- Semantic reasoning across multiple lectures
- Fully autonomous summarization engine

Product Features Not Included

- Mobile app (iOS/Android)
- Offline-first full LLM inference
- Collaborative shared notes
- Real-time quiz generation
- LMS integrations (Canvas/Blackboard)

Behavioral Guarantees Not Claimed

- Perfect Action Item detection
- Zero hallucinations
- Consistent performance across all lecture styles
- Stable performance under poor audio conditions